

Rishab Premod Nambiar

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SUMMARY

Machine Learning-focused Data Science student building end-to-end AI systems across NLP, Computer Vision, and Reinforcement Learning. Experienced with Transformers, GANs, XGBoost, and Explainable AI, with projects involving model deployment on real-world datasets.

EDUCATION

Amrita Vishwa Vidyapeetham

Integrated MSc Data Science (4th Year) — GPA: 8.99/10

Coimbatore, India

Sept 2022 – Present

Kendriya Vidyalaya No 1, Calicut

AISSCE — 95.2% — Physics, Chemistry, Mathematics, Biology

2020–21

TECHNICAL SKILLS

ML/DL & AI: PyTorch, Scikit-learn, XGBoost, Transformers (RoBERTa, BERT), CNNs, GANs, DQN, NLP, Computer Vision, Explainable AI (SHAP, LIME, Grad-CAM)

Programming: Python, R, SQL, MATLAB

Data & Visualization: NumPy, Pandas, SciPy, Matplotlib, Plotly, Tableau

Deployment: Streamlit, Gradio

PROJECTS

DetectAI: Human vs AI-Generated Text Classifier

- Built a hybrid detection system combining TF-IDF + Random Forest + BiLSTM with a fine-tuned RoBERTa transformer, achieving **98.64% accuracy**.
- Integrated SHAP and LIME for word-level interpretability; deployed an interactive demo via Gradio for real-time inference.

Image Colorization using GANs

- Developed an end-to-end Pix2Pix GAN pipeline to convert grayscale images to realistic color outputs (PSNR: 24.56, SSIM: 0.899).
- Designed reproducible preprocessing and evaluation workflows in Python for metric-based model comparison.

Adaptive Traffic Signal Controller using Reinforcement Learning

- Built a Deep Q-Network (DQN) agent that learned optimal traffic signal phase-switching, reducing congestion by **~10%** vs fixed-time baseline.
- Developed a PyGame simulation environment for iterative agent training and real-time policy visualization.

MockViewAI – AI-Powered Mock Interview System

- Generating role-specific interview questions from resumes/JDs and implementing automated response scoring with structured feedback.

Attention-Based Trimodal Fusion of Visual, Structural, and Semantic Features for Malware Image Classification

- Achieved **99.57%** accuracy for malware image classification using a trimodal fusion of CNN + GNN + Semantic Attributes

Predictive Demand Modeling – Logistics Industry

Hackathon Winner

- Built a time-series forecasting pipeline using XGBoost and engineered temporal features to predict grocery sales; **won SMILe Logistics Innovation Hack**.

Illegal Fence Detection for Wildlife Conservation (YOLOv4)

Published – ICAISC-2025

- Developed a drone-based AI system using YOLOv4 to detect illegal electric fences in wildlife areas; paper presented at ICAISC-2025.

CERTIFICATIONS & ACTIVITIES

Certifications: Python (HackerRank), Problem Solving (HackerRank)

Leadership: Technical team member, Ideathon '24 Hackathon at Anokha Tech Fest, Amrita Vishwa Vidyapeetham

Awards: 4th & 5th Prize, AIU Inter-University Youth Festival South Zone (Group Song, 38th & 39th editions)